



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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GitHub: <https://github.com/luannt0801/pcap-31-03-practice/tree/main/IBM/spacex-capstone>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies**

- Data collection with the SpaceX REST API and web scraping of Wikipedia (BeautifulSoup)
- Data wrangling: missing values, landing-outcome label (Class), one-hot encoding
- Exploratory data analysis with SQL (SQLite) and data visualization (Matplotlib / Seaborn)
- Interactive visual analytics with Folium maps and a Plotly Dash dashboard
- Predictive analysis: Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV

- **Summary of all results**

- 90 Falcon 9 launches analysed; overall first-stage landing success rate is 66.7%
- Success rate rose from 0% (2010–2013) to over 80% since 2017 as SpaceX gained experience
- KSC LC-39A has the most successful launches (41.7% of all successes, 76.9% success ratio)
- SSO, ES-L1, GEO and HEO orbits reached 100% success; GTO is the hardest frequent orbit (52%)
- Best model: SVM (sigmoid kernel) – 84.8% cross-validation accuracy and 83.3% test accuracy
- New insight: boosters with grid fins + landing legs land 83% of the time vs. 14% without them

Project at a Glance

90

Falcon 9 launches analysed
(2010–2020)

66.7%

overall first-stage landing
success

0% → 84%

success rate 2010–2013 vs.
2020

41.7%

of all successful launches
come from KSC LC-39A

83% vs 14%

landing success with vs.
without grid fins + legs

85%

landing success of Block 5
boosters

4 × GridSearchCV

LR, SVM, Decision Tree and
KNN tuned with 10-fold CV

83.3%

test accuracy – best model:
SVM (sigmoid kernel)

Introduction

- **Project background and context**

- SpaceX advertises Falcon 9 launches at about US\$62 million, while other providers charge US\$165 million or more.
- Most of the saving comes from recovering and reusing the first stage after it lands.
- If we can predict whether the first stage will land, we can estimate the cost of a launch. This helps a new company (Space Y) bid against SpaceX.

- **Problems we want to find answers to**

- Which factors (payload mass, orbit, launch site, flight number, booster) affect a successful landing?
- How has the landing success rate changed over the years?
- Where are the launch sites located, and what is around them?
- Which classification model predicts the landing outcome best?

Section 1

Methodology

Methodology

Executive Summary

Data collection methodology:

- SpaceX REST API (launch, rocket, launchpad, payload and core data) and web scraping of the Wikipedia Falcon 9 launch tables

Perform data wrangling

- Kept Falcon 9 launches, filled missing payload mass with the mean, created the Class label (1 = landed, 0 = failed) and one-hot encoded categorical features

Perform exploratory data analysis (EDA) using visualization and SQL

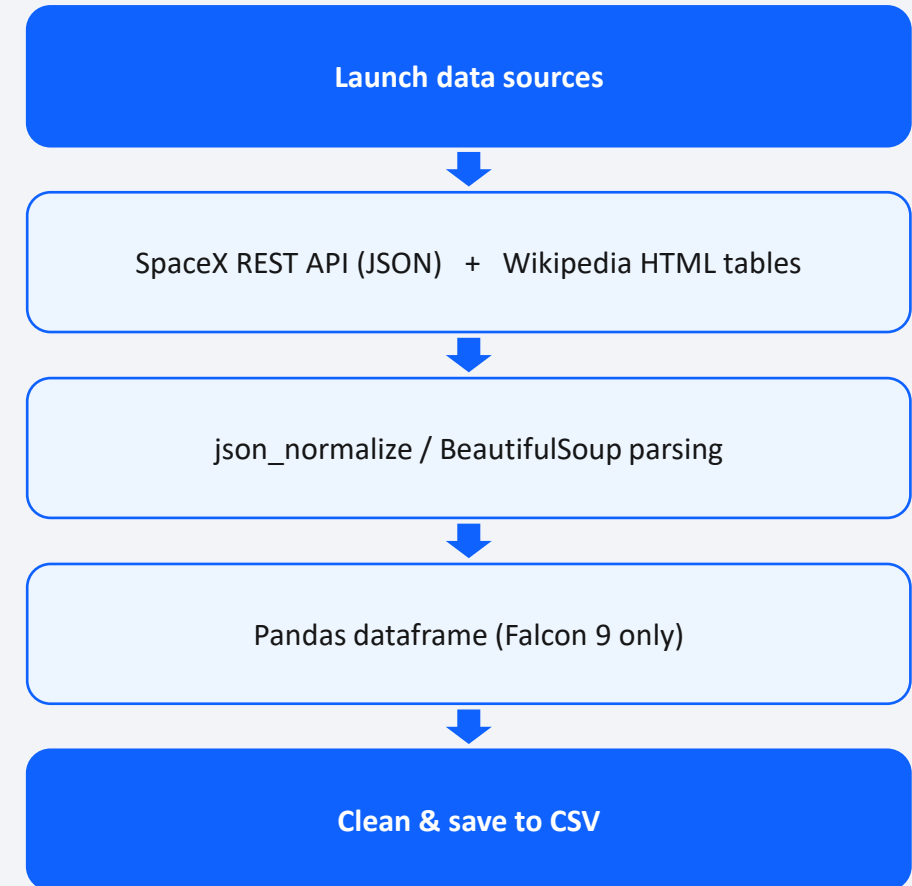
Perform interactive visual analytics using Folium and Plotly Dash

Perform predictive analysis using classification models

- Standardized the features, split 80/20 into train/test, tuned LR, SVM, Decision Tree and KNN with GridSearchCV (10-fold CV) and compared their accuracy on the test set

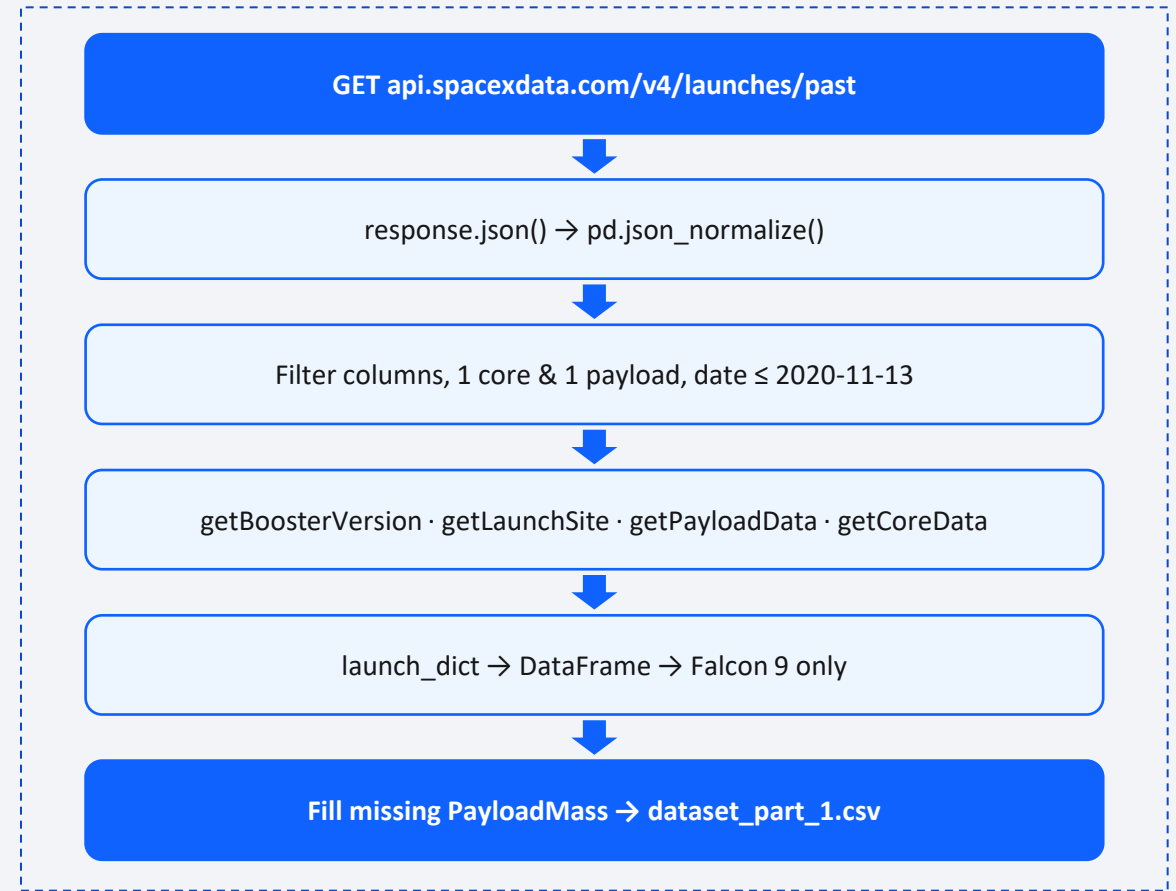
Data Collection

- Two data sources were combined:
 - SpaceX REST API (api.spacexdata.com/v4): past launches with IDs for rocket, launchpad, payloads and cores; extra GET requests return booster version, site coordinates, payload mass, orbit and landing outcome.
 - Web scraping: the Wikipedia page 'List of Falcon 9 and Falcon Heavy launches' was parsed with BeautifulSoup to extract the launch tables.
- The JSON and HTML were normalised into Pandas dataframes, filtered to Falcon 9 launches and saved to CSV for wrangling.



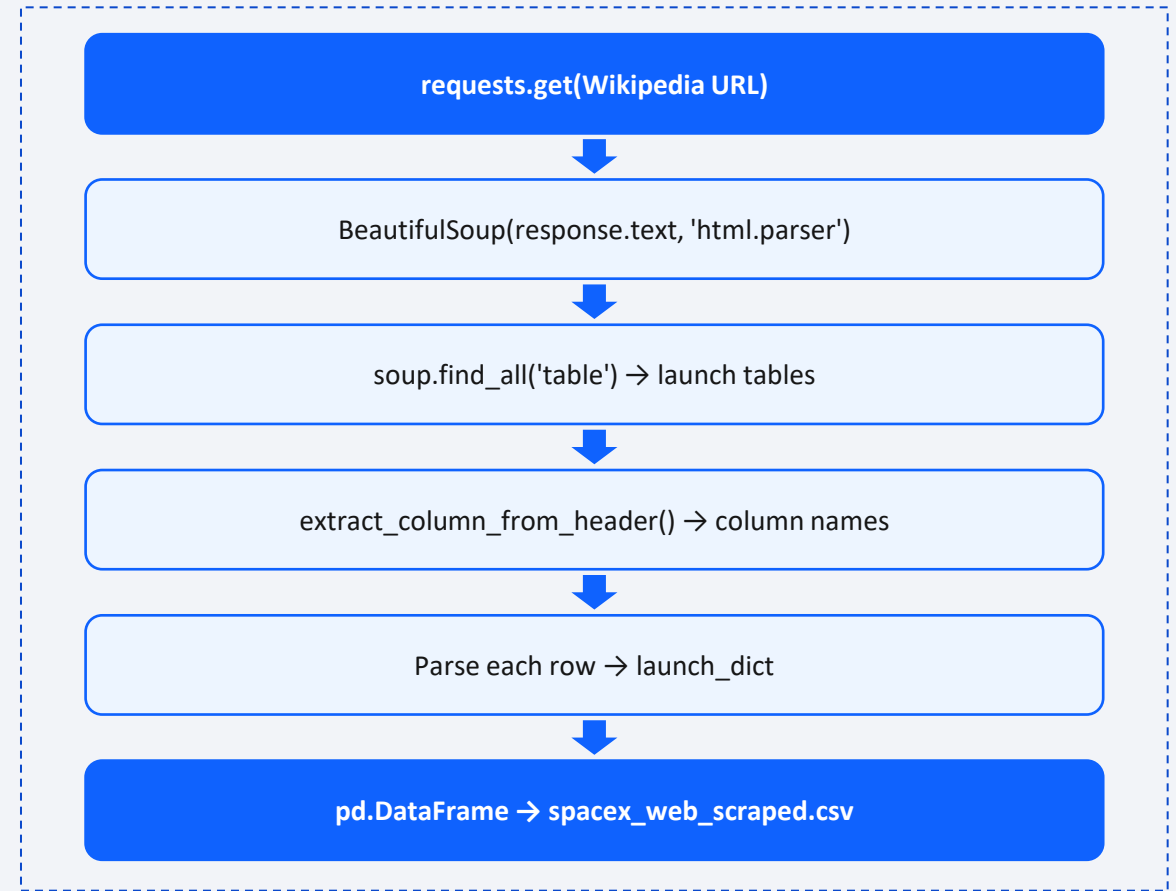
Data Collection – SpaceX API

- GET request to /v4/launches/past and decode the JSON response
- `pd.json_normalize()` to flatten the JSON into a dataframe
- Keep rocket, payloads, launchpad, cores, flight number and date; drop multi-core / multi-payload launches
- Call /rockets, /launchpads, /payloads and /cores for each ID to get the details
- Build a dictionary, keep only Falcon 9 launches (90 rows), replace missing PayloadMass with the mean
- **GitHub:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/1_data_collection_api.ipynb



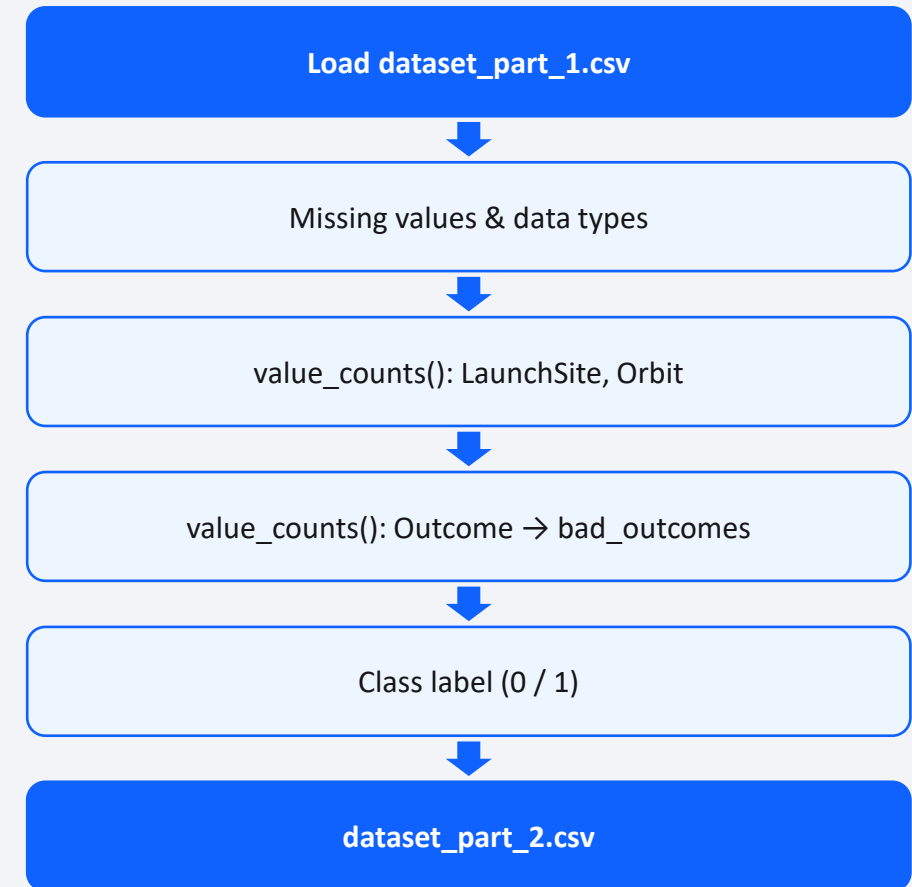
Data Collection - Scraping

- Request the Falcon 9 launch Wikipedia page (snapshot 9 June 2021) with `requests.get()`
- Create a BeautifulSoup object from the HTML response
- Find all tables and extract the column names from the header of the launch table
- Iterate over the rows of every 'wikitable' to collect flight number, date, booster, site, payload, orbit, customer and outcomes (121 rows)
- Convert the dictionary to a dataframe and export to CSV
- **GitHub:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/2_webscraping.ipynb



Data Wrangling

- Explored the dataset: missing values per column and data types
- Counted the launches per launch site and the occurrence of each orbit
- Counted the landing outcomes (e.g. True ASDS, False Ocean, None None)
- Converted the outcomes into a training label Class: 1 = first stage landed, 0 = did not land (overall success rate 66.7%)
- Exported the labelled data to dataset_part_2.csv
- **GitHub:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/3_data_wrangling.ipynb



EDA with Data Visualization

- **Scatter (cat) plots** – Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload vs. Launch Site, Flight Number vs. Orbit, Payload vs. Orbit (coloured by Class): show how two variables relate and where successful landings cluster
- **Bar chart** – success rate of each orbit type: compares a numeric value across categories
- **Line chart** – average success rate per year: shows the trend over time
- **Feature engineering** – selected 12 features and one-hot encoded Orbit, LaunchSite, LandingPad and Serial (80 columns) for the models
- **GitHub:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/5_eda_visualization.ipynb

EDA with SQL

- Loaded SpaceX.csv into a SQLite database (table SPACEXTABLE) and queried it with SQL:
 - Names of the unique launch sites (SELECT DISTINCT)
 - 5 records where the launch site begins with 'CCA' (LIKE 'CCA%' LIMIT 5)
 - Total payload mass carried by boosters launched by NASA (CRS) (SUM)
 - Average payload mass carried by booster version F9 v1.1 (AVG)
 - Date of the first successful landing on a ground pad (MIN)
 - Boosters with a successful drone-ship landing and payload between 4000 and 6000 kg
 - Total number of successful and failed mission outcomes (GROUP BY)
 - Boosters that carried the maximum payload mass (subquery with MAX)
 - Failed drone-ship landings in 2015 with booster version and launch site (substr on Date)
 - Ranking of landing outcomes between 2010-06-04 and 2017-03-20 (GROUP BY, ORDER BY DESC)
- GitHub: https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/4_eda_sql.ipynb

Build an Interactive Map with Folium

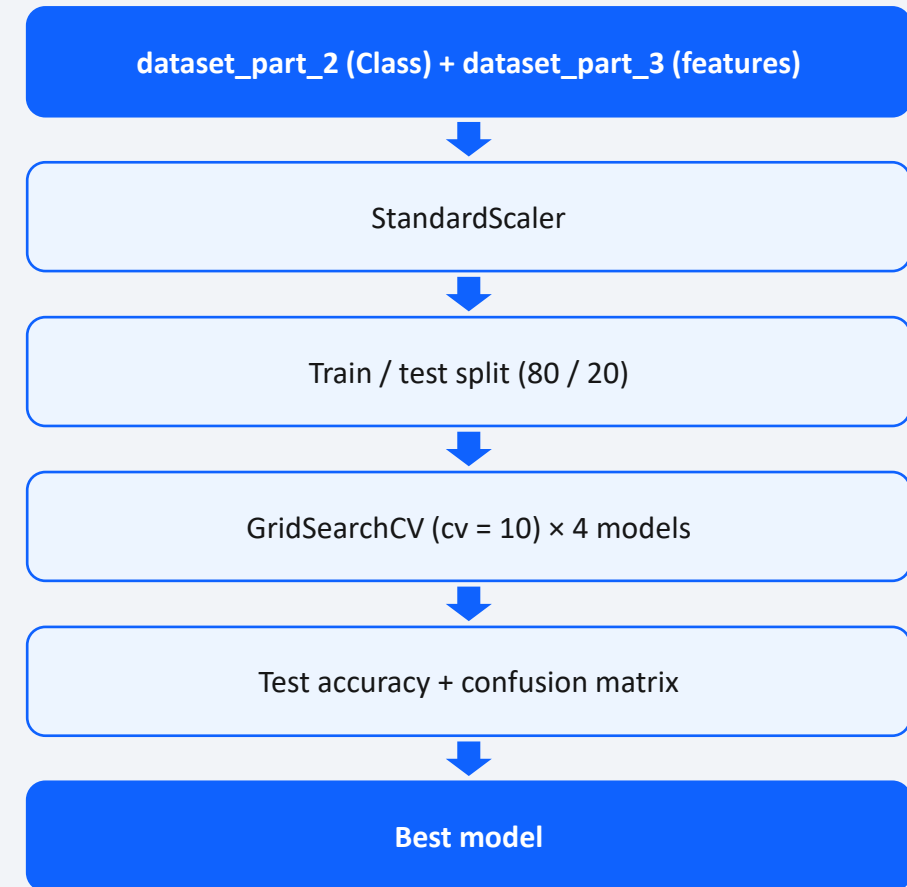
- **folium.Circle** + **folium.Marker** (**DivIcon** label) at each launch site – to see where the sites are (near the equator and the coast)
- **MarkerCluster** of green / red markers for successful / failed launches – many launches share the same coordinates, the cluster keeps the map readable and shows the success ratio per site
- **MousePosition** plugin – to read the coordinates of nearby points of interest
- **PolyLine** + **distance labels** from CCAFS SLC-40 to the nearest coastline, highway, railway and city – to analyse the proximities of a launch site (distances with the Haversine formula)
- **GitHub:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/6_folium_launch_site_locations.ipynb

Build a Dashboard with Plotly Dash

- **Dropdown (launch site)** – choose 'All Sites' or one site; drives both charts
- **Pie chart** – total successful launches by site (All Sites) or success vs. failure for the selected site: quickly compares the performance of the sites
- **Range slider (payload mass 0–10,000 kg)** – filter launches by payload
- **Scatter plot** – payload mass vs. launch outcome, coloured by booster version: shows which payload ranges and booster versions succeed most
- The callbacks update the charts instantly, so stakeholders can explore the data without code.
- **GitHub:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/7_spacex_dash_app.py

Predictive Analysis (Classification)

- **Build:** Y = Class; X = 80 engineered features, standardized with StandardScaler; train_test_split 80/20 (random_state = 2, 18 test samples)
- **Tune:** GridSearchCV (cv = 10) for Logistic Regression, SVM, Decision Tree and KNN
- **Evaluate:** best cross-validation accuracy, accuracy on the test set and confusion matrix for each model
- **Select:** the model with the highest accuracy
- **GitHub:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/8_machine_learning_prediction.ipynb



Results

- **Exploratory data analysis results**

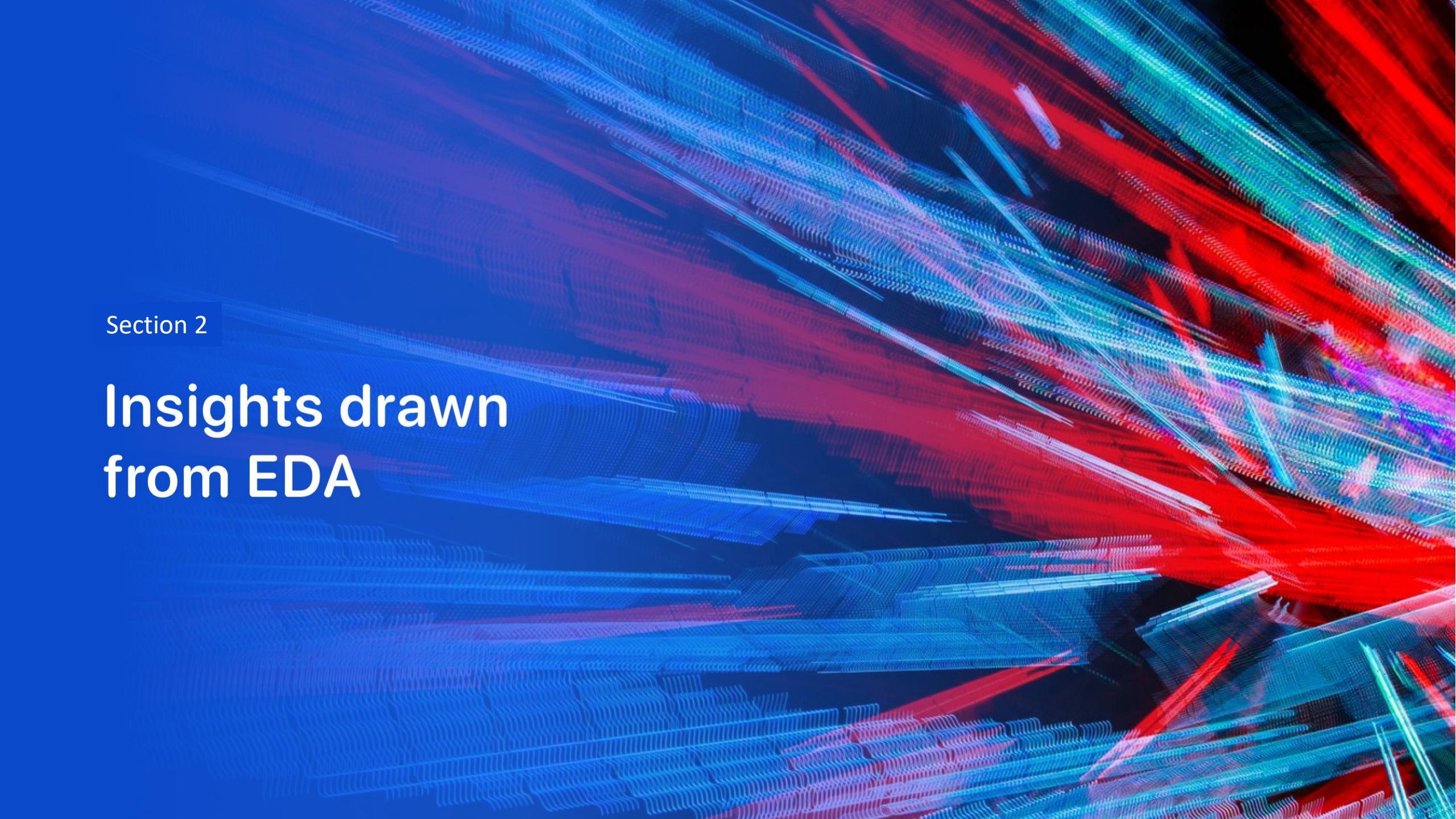
- Landing success improved over time and with flight number; KSC LC-39A and VAFB SLC-4E have the highest success ratios; SSO/ES-L1/GEO/HEO orbits have 100% success

- **Interactive analytics demo in screenshots**

- Folium maps of the launch sites, the launch outcomes and the proximities, and the Plotly Dash dashboard (Sections 3 and 4)

- **Predictive analysis results**

- All models reach 83.3% accuracy on the test set; SVM, Decision Tree and KNN have the best cross-validation accuracy (84.8%)

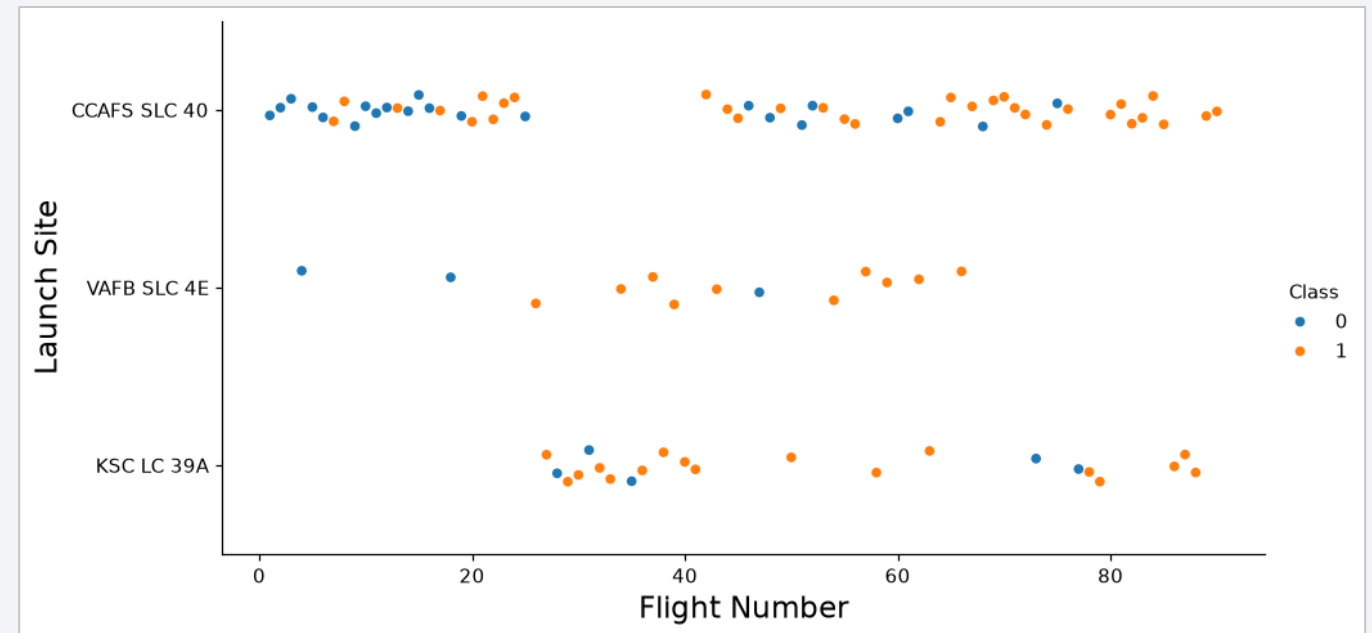
The background of the slide is an abstract composition. It features a dark blue gradient on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

Insights drawn from EDA

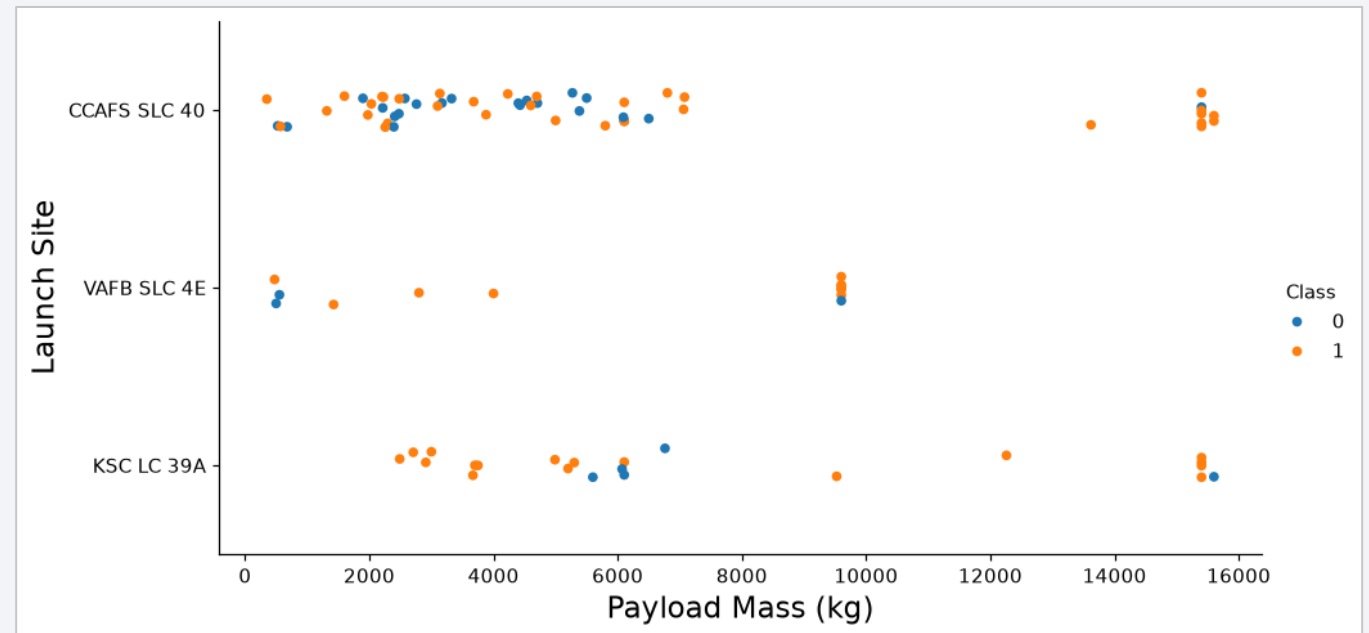
Flight Number vs. Launch Site

- **Why this chart:** a scatter (cat) plot shows every launch as a point, so we can see how the outcome changes with the flight number at each site.
- Most early flights were launched from CCAFS SLC 40 and most of them failed.
- As the flight number increases, the landing success rate increases at every site.
- CCAFS SLC 40 has the most launches (55); KSC LC 39A and VAFB SLC 4E have higher success ratios (~77%).



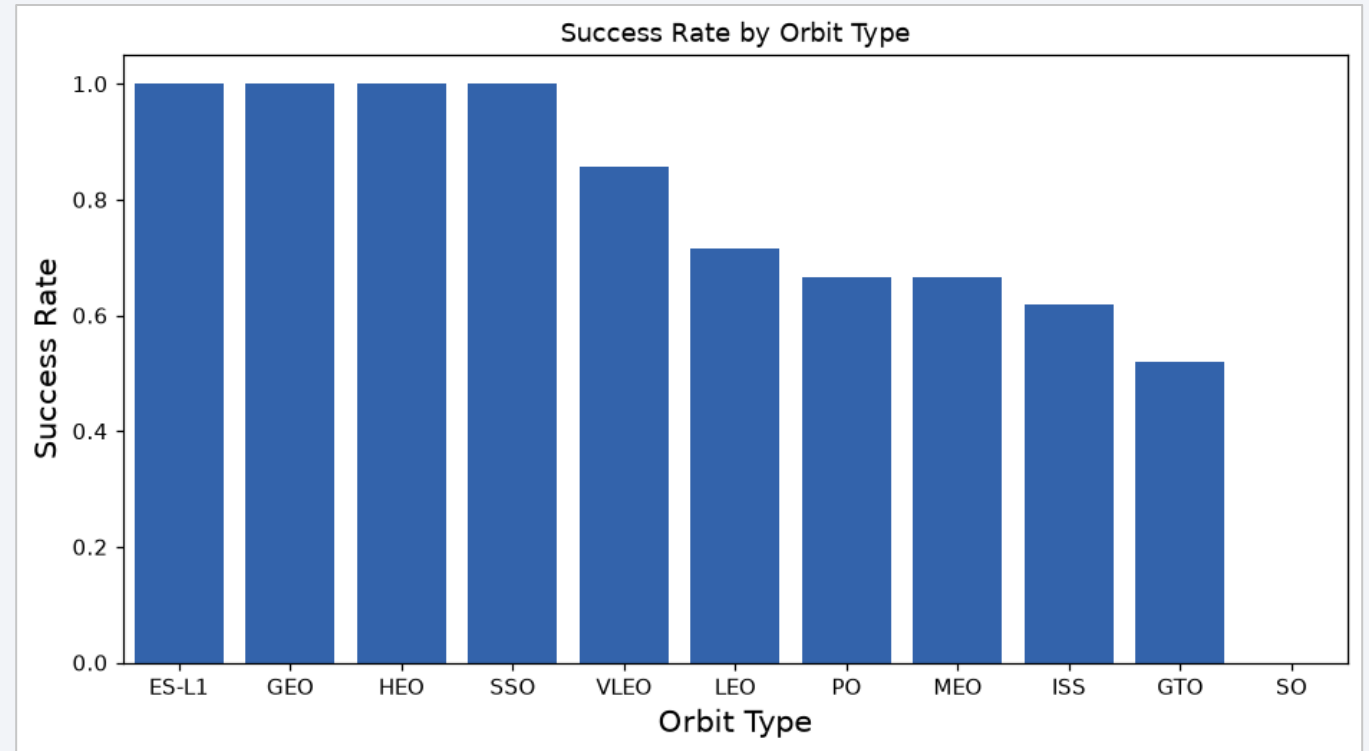
Payload vs. Launch Site

- **Why this chart:** a scatter plot relates a numeric variable (payload mass) to a categorical one (launch site), coloured by outcome.
- VAFB SLC 4E has no launches with a payload heavier than ~10,000 kg.
- Heavy payloads (> 7,000 kg) have a high success rate (88%), mostly from CCAFS SLC 40 and KSC LC 39A.
- For light payloads the outcome is more mixed.



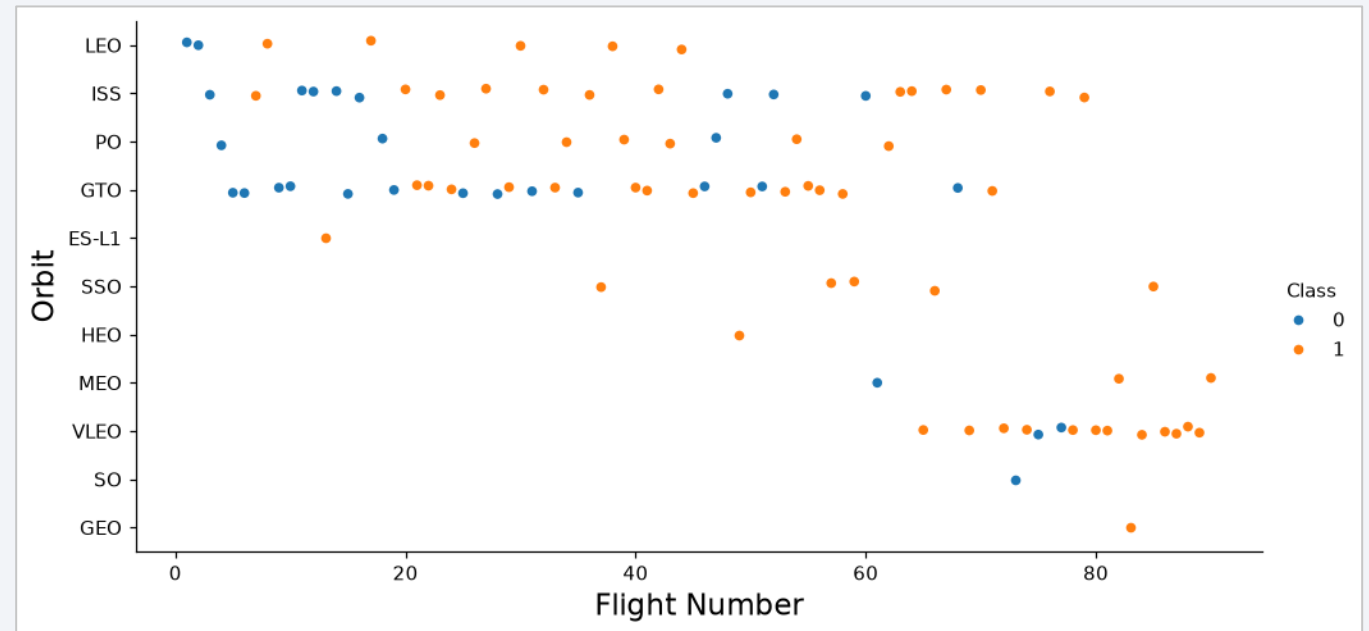
Success Rate vs. Orbit Type

- **Why this chart:** a bar chart is the clearest way to compare one numeric value (success rate) across categories (orbits).
- ES-L1, GEO, HEO and SSO have a 100% success rate (but only 1–5 launches each).
- VLEO (86%) and LEO (71%) are also high.
- GTO has the lowest success rate among frequent orbits (52%); SO had one failed launch.



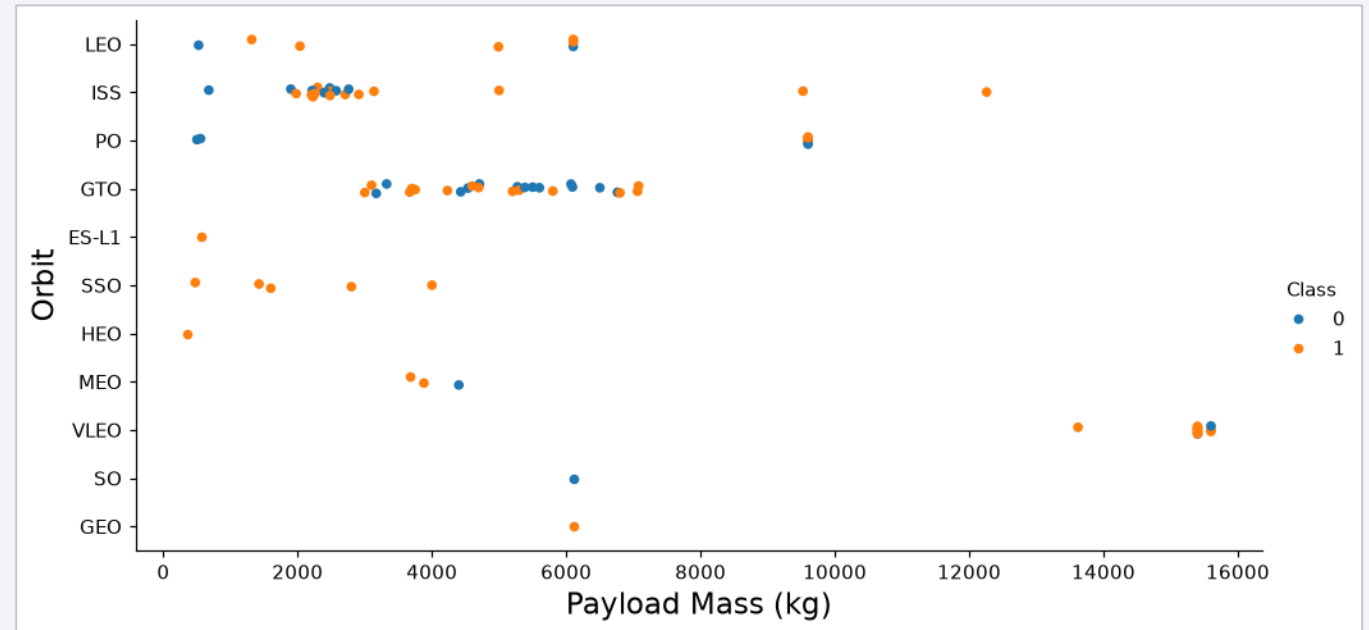
Flight Number vs. Orbit Type

- **Why this chart:** a scatter plot shows whether orbit choice and experience (flight number) together affect the outcome.
- In LEO, success is related to the number of flights: later flights succeed.
- In GTO there is no clear relationship with flight number.
- Recent flights move to VLEO (Starlink) with a high success rate.



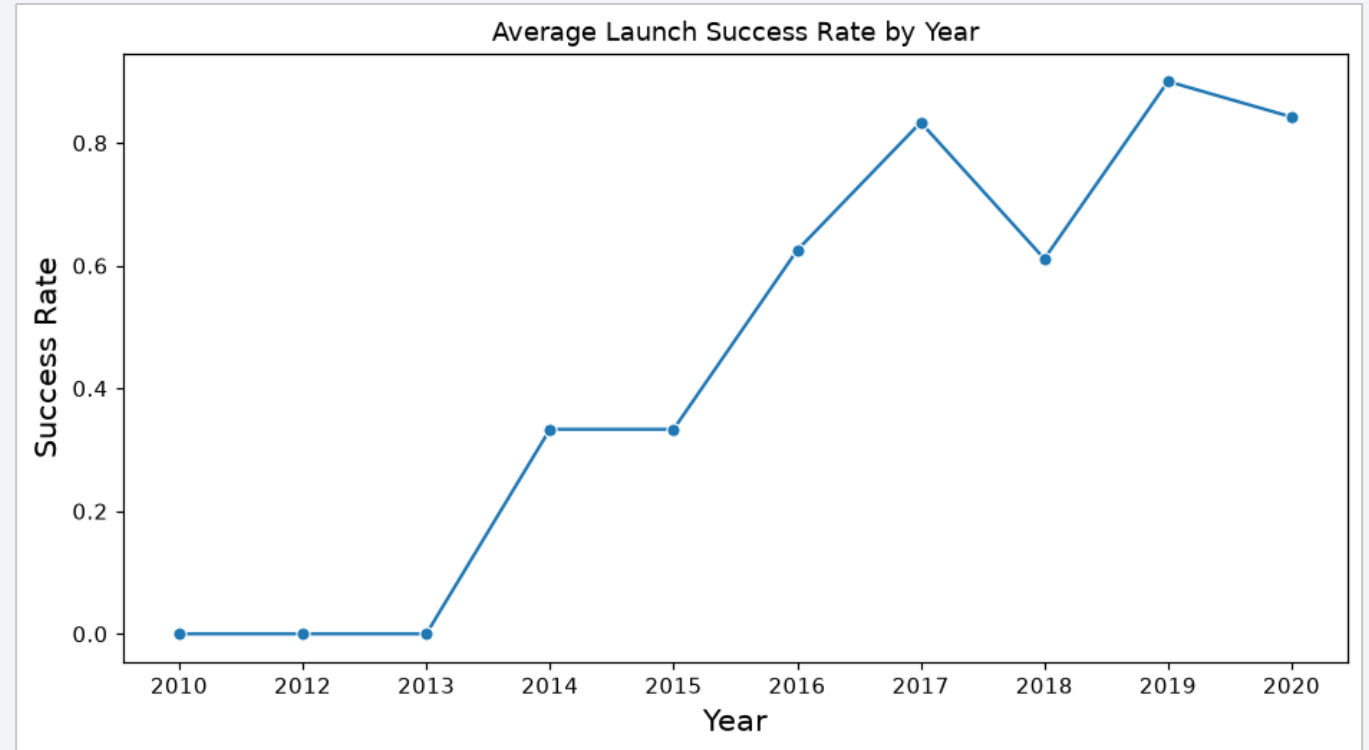
Payload vs. Orbit Type

- **Why this chart:** a scatter plot shows the payload range used for each orbit and where failures happen.
- Heavy payloads have a higher success rate for Polar, LEO and ISS orbits.
- For GTO, successful and failed landings are mixed across all payload masses.
- VLEO launches carry the heaviest payloads (~15,600 kg) and mostly land.



Launch Success Yearly Trend

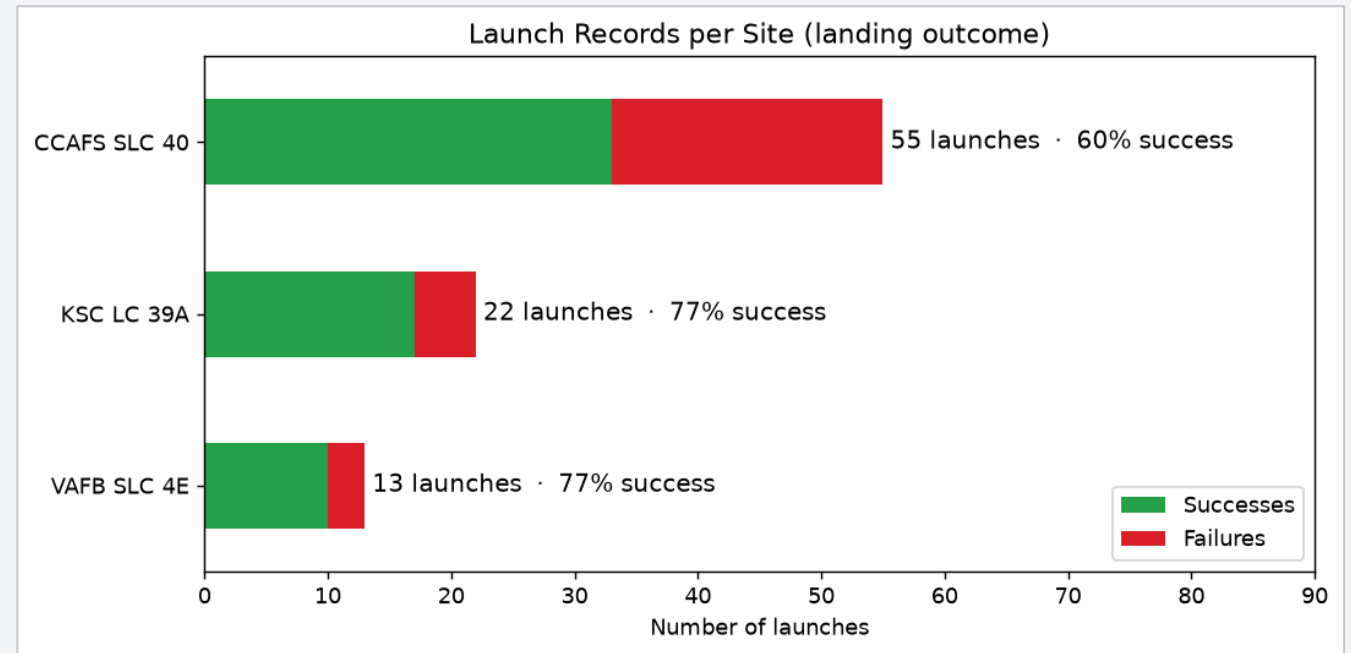
- **Why this chart:** a line chart is the best choice to show a trend over time (year by year).
- The success rate was 0% from 2010 to 2013.
- It kept increasing from 2013 to 2020 (small dip in 2018).
- Since 2017 the success rate stays above 60%, reaching 90% in 2019 and 84% in 2020.



Launch Records per Site

- **Why this chart:** a stacked horizontal bar compares the number of launches per site and splits each bar into successful and failed landings.
- CCAFS SLC 40 (includes the former LC-40 pad) has the most launches (55) but the lowest success rate (60%).
- KSC LC 39A (77%) and VAFB SLC 4E (77%) have fewer launches but a higher success rate.

Launch Site	Launches	Landed	Rate
CCAFS SLC 40	55	33	60.0%
KSC LC 39A	22	17	77.3%
VAFB SLC 4E	13	10	76.9%



All Launch Site Names

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- DISTINCT returns each launch site only once.
- SpaceX used 4 launch sites: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A (Florida) and VAFB SLC-4E (California).

Launch Site Names Begin with 'CCA'

```
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualificat	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two C	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- LIKE 'CCA%' matches site names starting with 'CCA'; LIMIT 5 returns the first 5 records.
- All 5 are early F9 v1.0 launches from CCAFS LC-40 (2010–2013) to LEO; none attempted or achieved a landing.

Total Payload Mass

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

Total_Payload_Mass
45,596 kg

- SUM adds the payload mass of all launches whose customer is NASA (CRS).
- Boosters launched for NASA (CRS) carried a total of 45,596 kg.

Average Payload Mass by F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

Avg_Payload_Mass
2,928.4 kg

- AVG computes the mean payload mass of launches with booster version F9 v1.1.
- The F9 v1.1 booster carried on average 2,928.4 kg.

First Successful Ground Landing Date

```
SELECT MIN(Date) AS First_Successful_Ground_Landing FROM SPACEXTABLE  
WHERE Landing_Outcome = 'Success (ground pad)';
```

First_Successful_Ground_Landing
2015-12-22

- MIN(Date) returns the earliest date with a successful ground-pad landing.
- The first successful ground-pad landing happened on 22 December 2015.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- WHERE combines the landing outcome with the payload range using AND.
- Four F9 FT boosters landed successfully on a drone ship with a 4,000–6,000 kg payload.

Total Number of Successful and Failure Mission Outcomes

```
SELECT Mission_Outcome, COUNT(*) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- GROUP BY counts the launches for each mission outcome.
- 100 missions succeeded (98 + 1 + 1 with unclear payload status) and only 1 failed in flight.

Boosters Carried Maximum Payload

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Booster_Version (1–6)	Booster_Version (7–12)
F9 B5 B1048.4	F9 B5 B1049.5
F9 B5 B1049.4	F9 B5 B1060.2
F9 B5 B1051.3	F9 B5 B1058.3
F9 B5 B1056.4	F9 B5 B1051.6
F9 B5 B1048.5	F9 B5 B1060.3
F9 B5 B1051.4	F9 B5 B1049.7

- A subquery finds the maximum payload mass; the outer query lists the boosters with that payload.
- 12 Falcon 9 Block 5 (F9 B5) boosters carried the maximum payload of 15,600 kg.

2015 Launch Records

```
SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE
WHERE substr(Date, 0, 5) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- substr() extracts the year and the month from the Date column.
- In 2015 there were 2 failed drone-ship landings (January and April), both F9 v1.1 boosters from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'  
GROUP BY Landing_Outcome ORDER BY Count DESC;
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

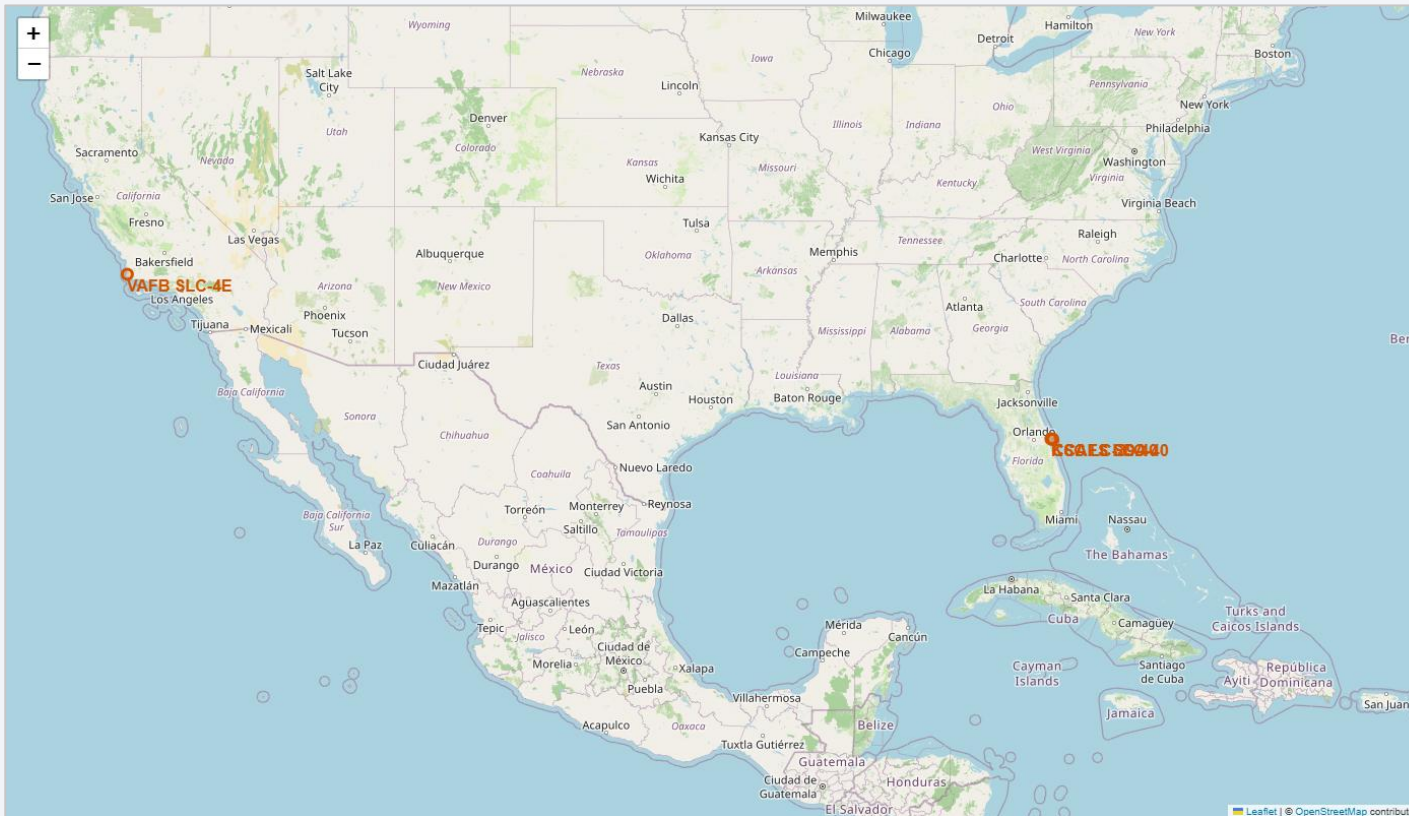
- BETWEEN filters the date range; GROUP BY + ORDER BY DESC rank the outcomes.
- 'No attempt' is the most frequent outcome (10). Drone-ship landings had 5 successes and 5 failures in this period.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

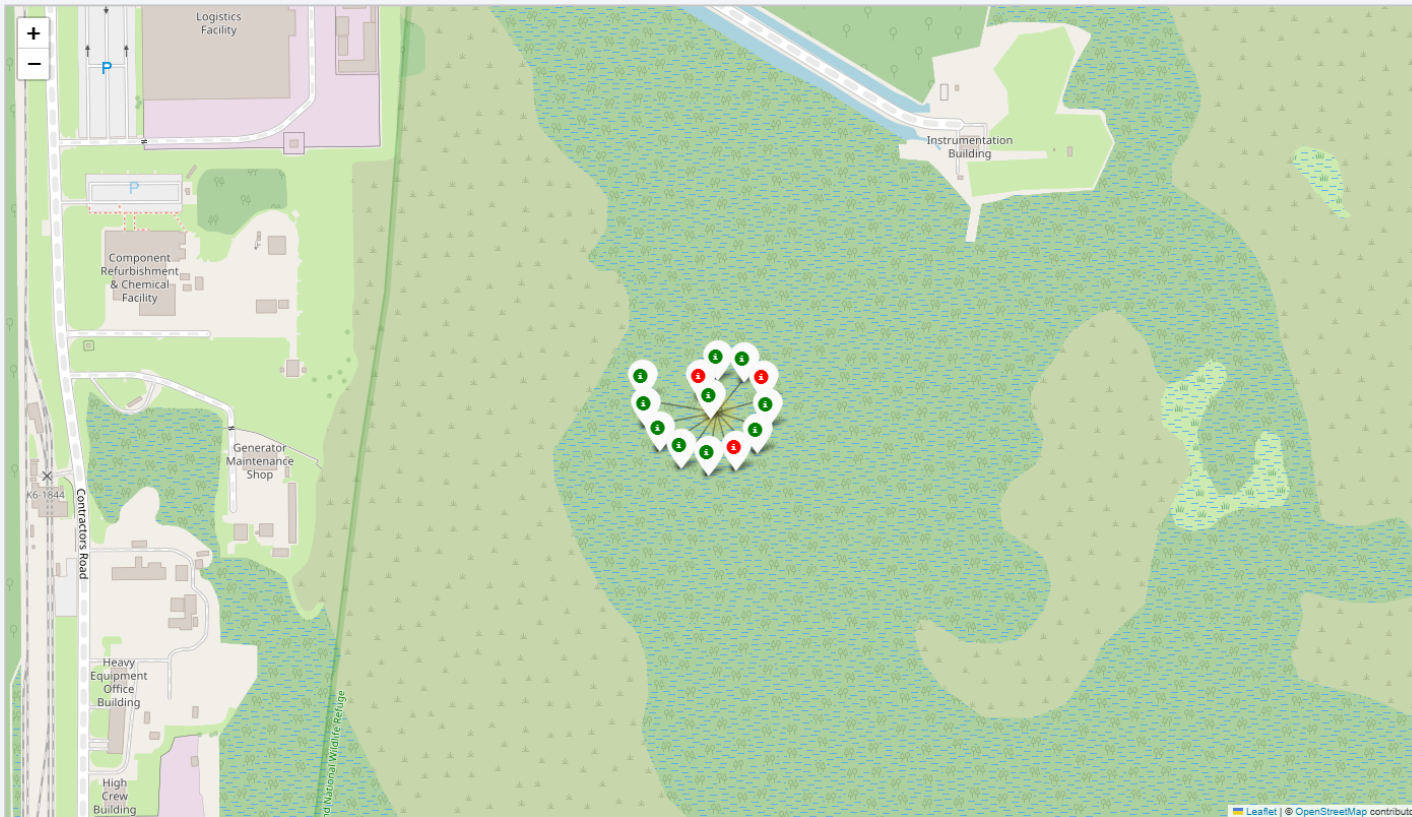
Launch Sites Proximities Analysis

All Launch Sites on the Global Map



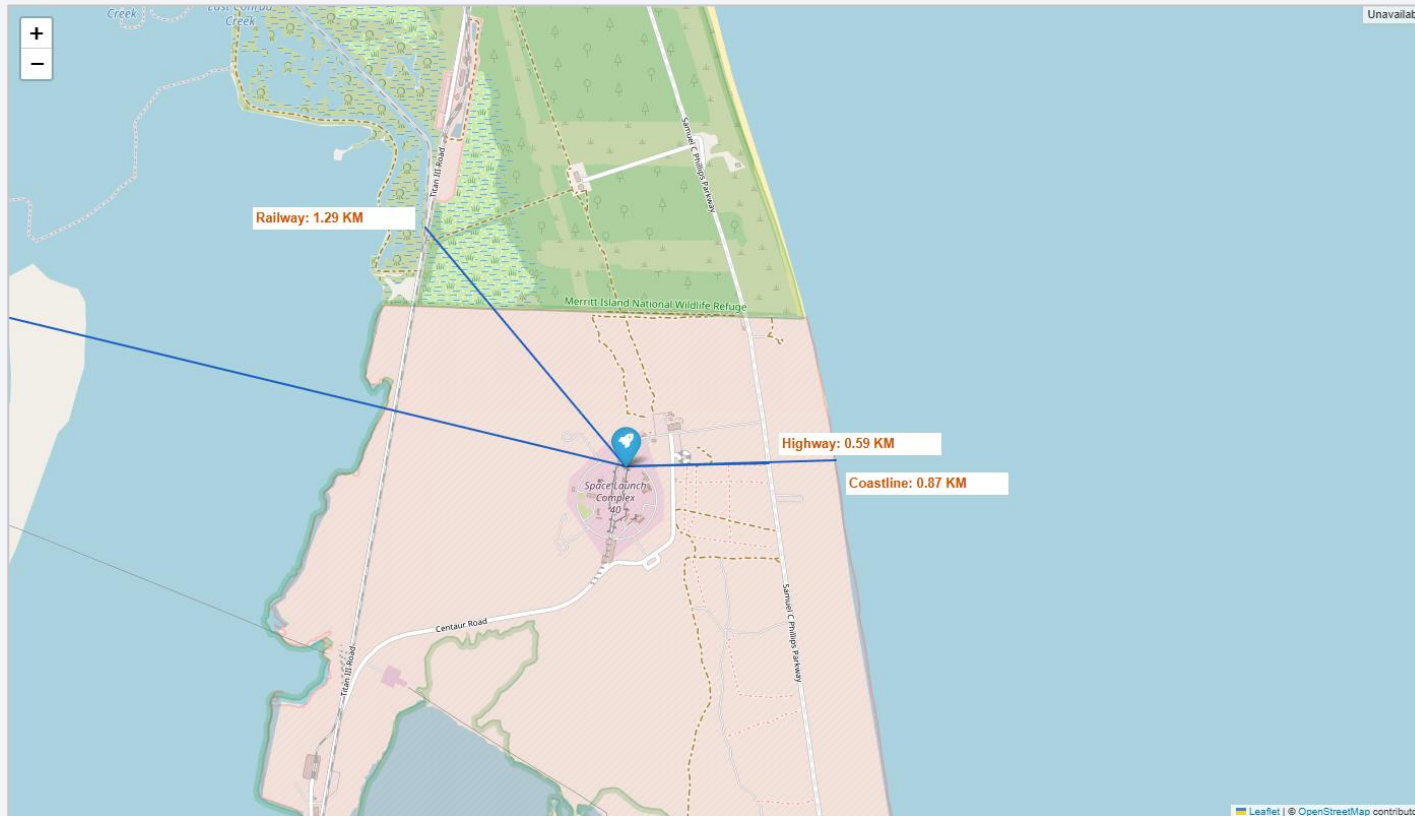
- Each launch site is marked with a circle and a name label.
- The sites are in the USA: 3 in Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and 1 in California (VAFB SLC-4E).
- All sites are close to the coast and as close to the equator as possible in the US, which helps launches and keeps rocket debris over the ocean.

Launch Outcomes by Site (Color-Labeled Markers)

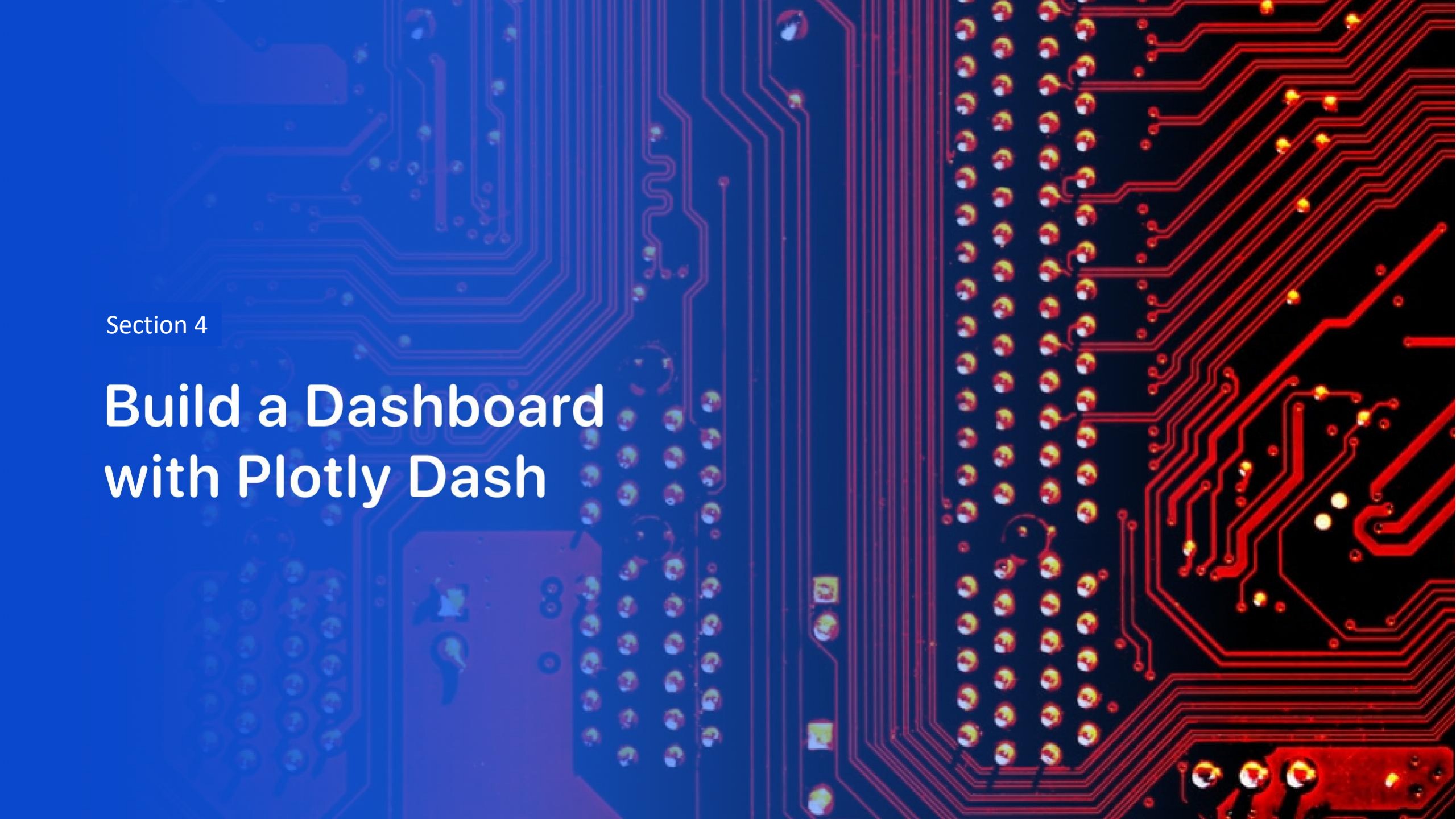


- Launches from the same site are grouped with a MarkerCluster; clicking the cluster shows each launch.
- Green marker = successful landing, red marker = failed landing.
- The screenshot shows KSC LC-39A: most of its markers are green, matching its high success ratio (10 of 13 launches).

Proximities of Launch Site CCAFS SLC-40



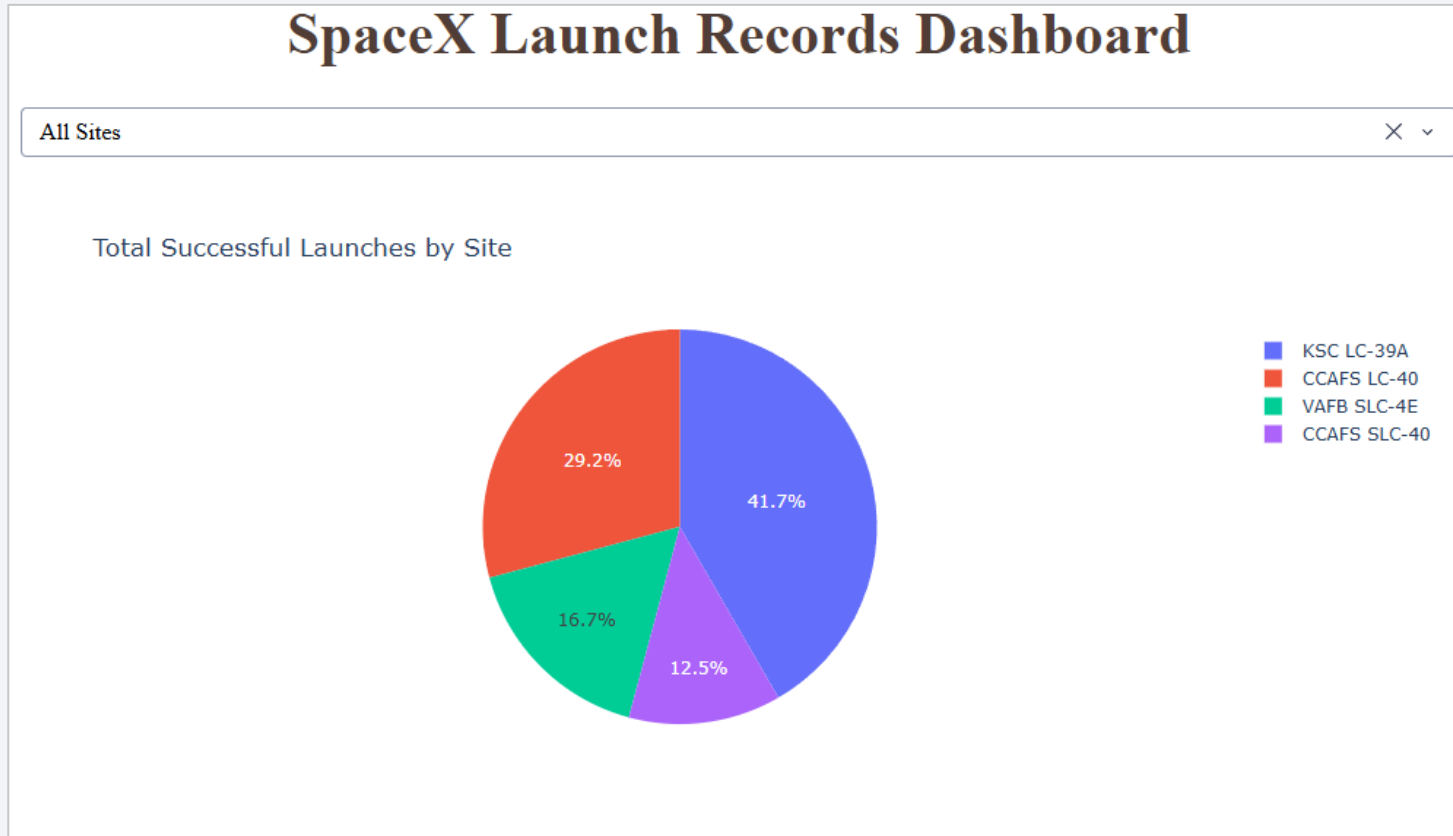
- Lines and distance labels from CCAFS SLC-40 to its closest points of interest:
 - Highway (Samuel C Phillips Pkwy): 0.59 km
 - Coastline: 0.87 km
 - Railway: 1.29 km
 - City (Titusville): 23.19 km
- Launch sites are close to the coast, highways and railways (transport of heavy parts) but keep a safe distance from cities.



Section 4

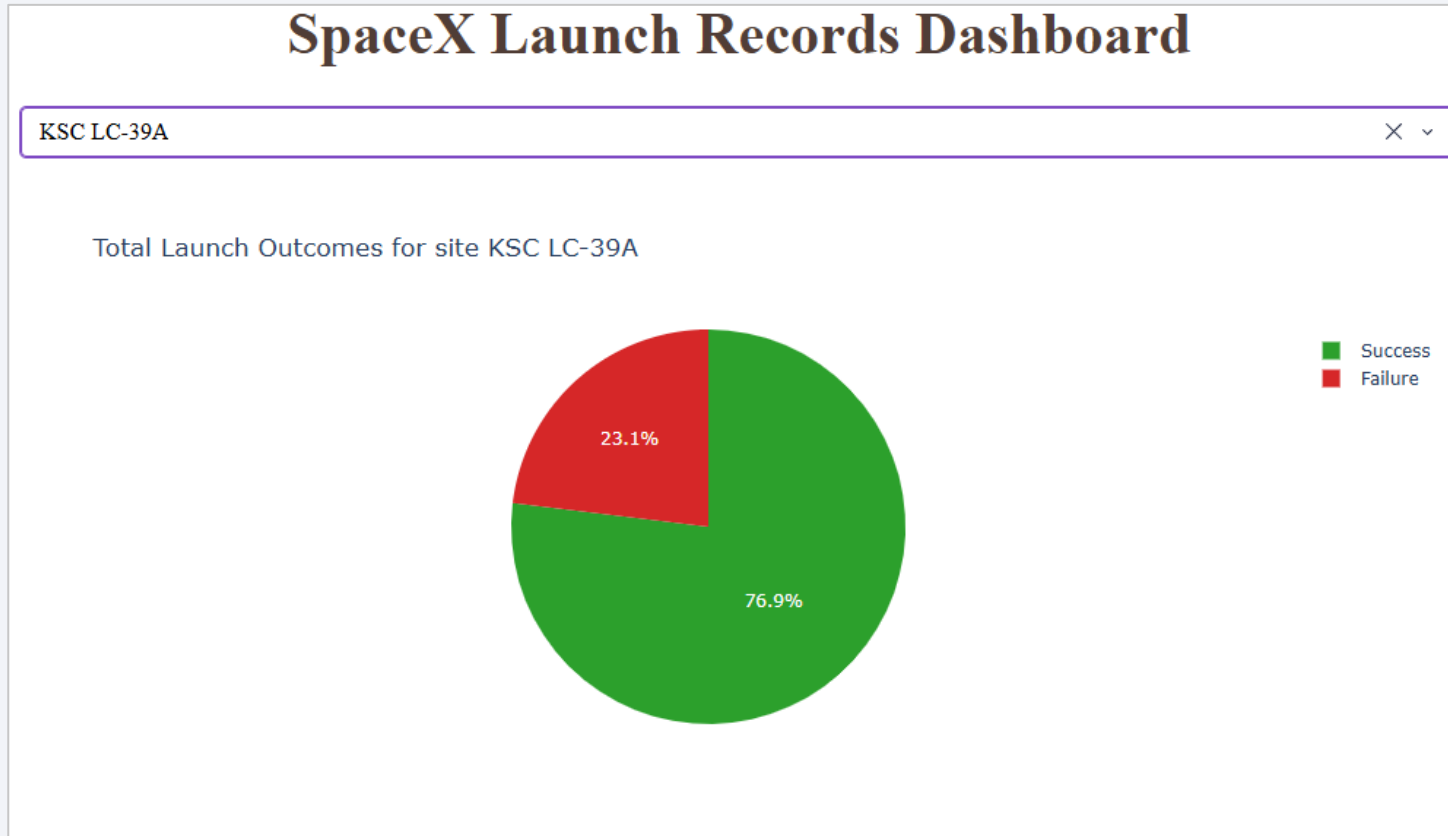
Build a Dashboard with Plotly Dash

Total Successful Launches by Site



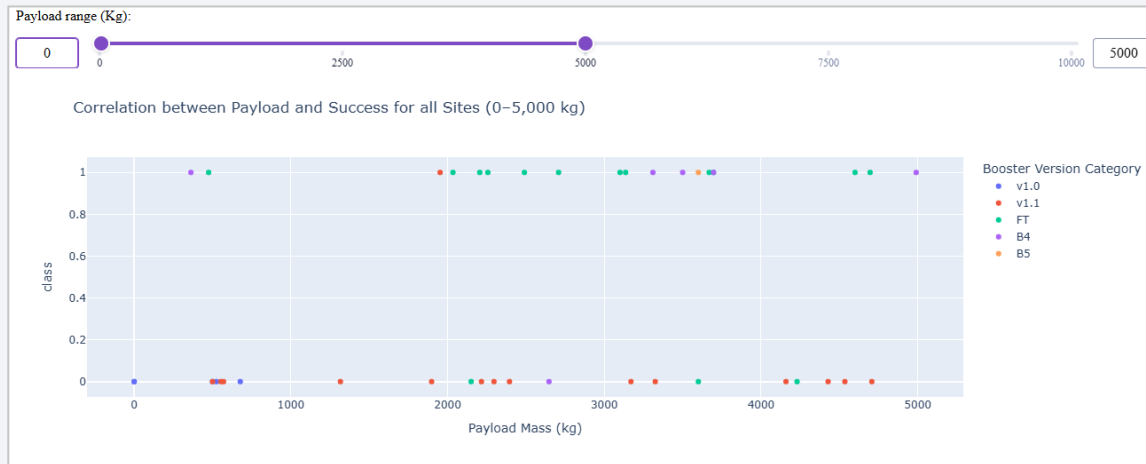
- Dashboard with the launch-site dropdown set to 'All Sites'; the pie chart shows how the 24 successful launches are shared between the sites.
- KSC LC-39A has the largest share of successful launches (41.7%, 10 launches).
- It is followed by CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%).
- Florida sites together produce 83% of all successful launches.

KSC LC-39A – Highest Launch Success Ratio



- Selecting one site in the dropdown switches the pie chart to success vs. failure for that site.
- KSC LC-39A has the highest success ratio: 76.9% of its launches were successful (10 of 13) and 23.1% failed (3 of 13).
- Success ratio of the other sites: CCAFS SLC-40 43% (3/7), VAFB SLC-4E 40% (4/10), CCAFS LC-40 27% (7/26).

Payload vs. Launch Outcome for All Sites



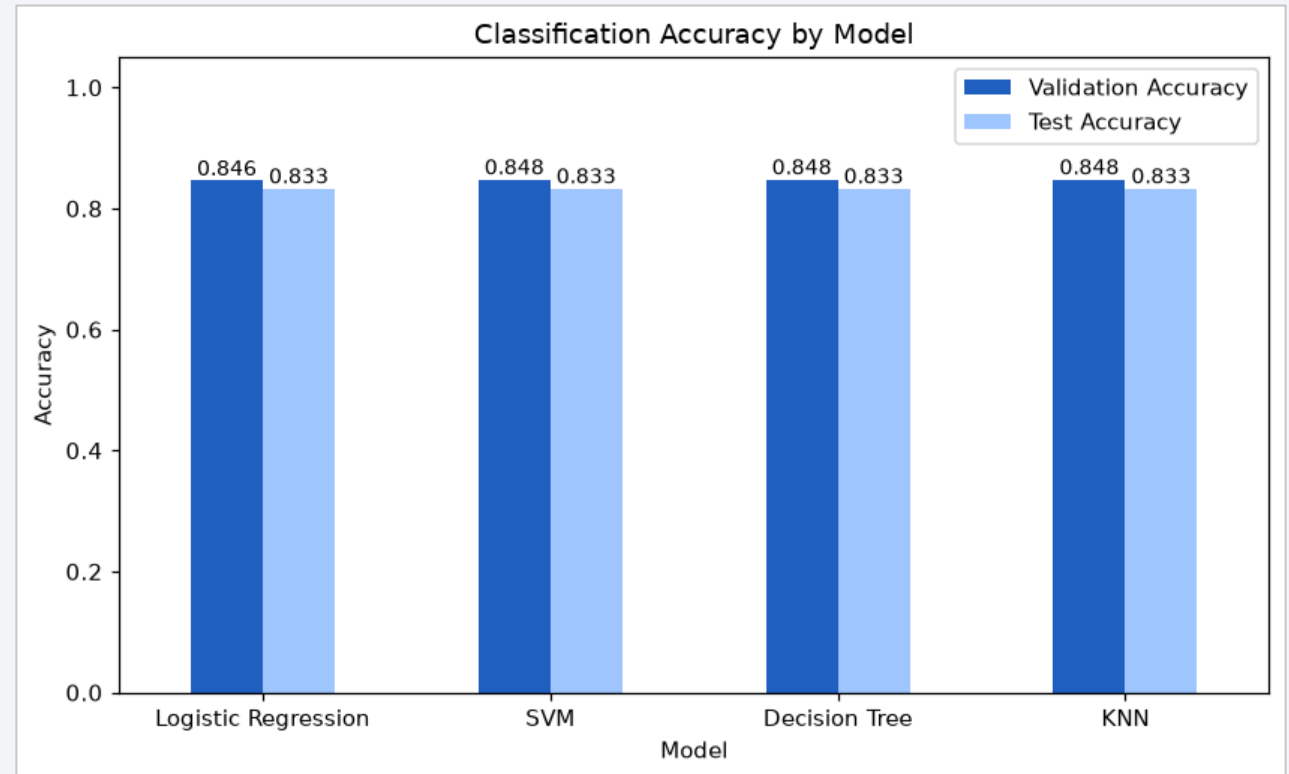
- The range slider filters the scatter plot by payload mass; the color shows the booster version category.
- **Payload 0 – 5,000 kg (top):** success rate 45%
- **Payload 5,000 – 10,000 kg (bottom):** success rate 36%
- Low payloads have a higher success rate than heavy payloads.
- Booster FT has the most successful launches (80% success below 5,000 kg); B4 also does well on light payloads. v1.0 and v1.1 boosters almost always failed to land.

Section 5

Predictive Analysis (Classification)

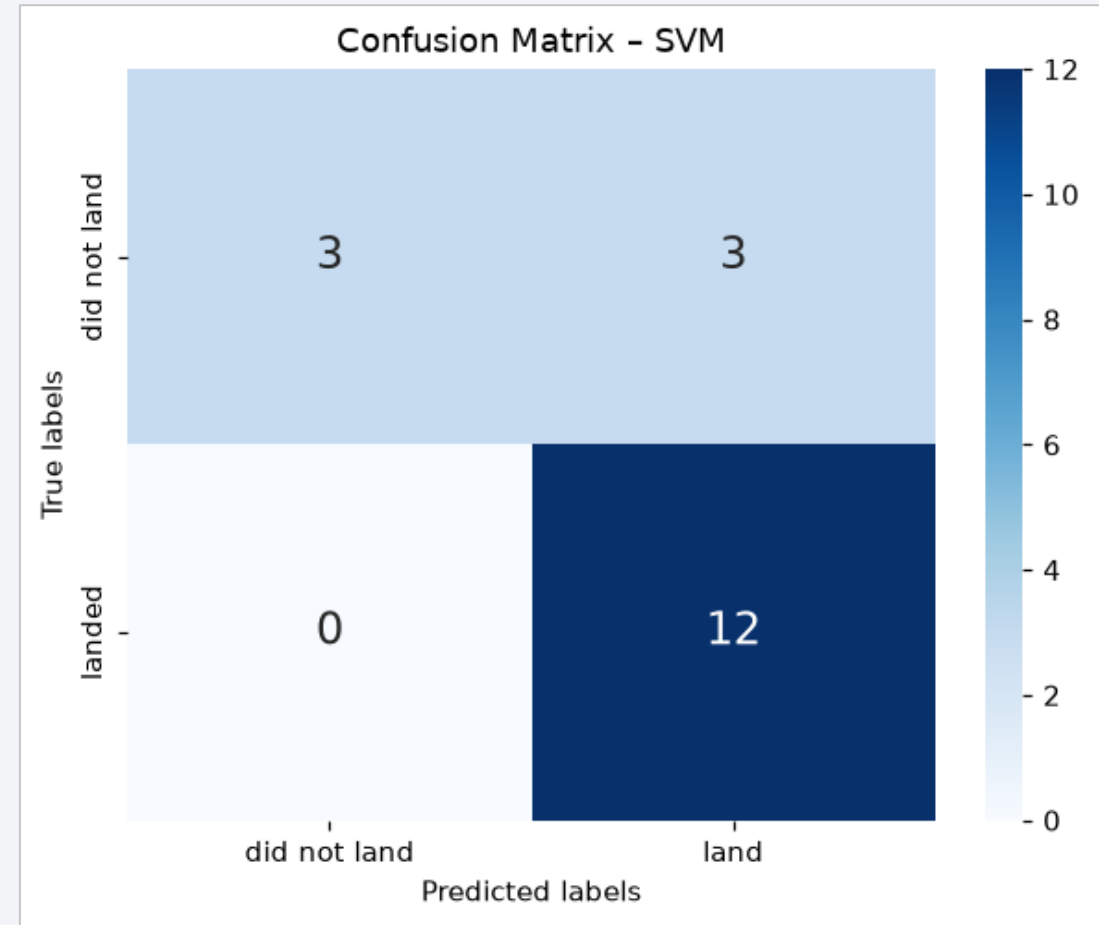
Classification Accuracy

- Bar chart of the cross-validation (GridSearchCV) and test accuracy of the four models.
- All models reach the same test accuracy of 83.3% (15 of 18 test samples).
- SVM (kernel = sigmoid), Decision Tree and KNN tie on the best validation accuracy (84.8%); Logistic Regression is slightly lower (84.6%).
- **Best model:** SVM (sigmoid kernel, C = 1) – highest validation accuracy (84.8%) and 83.3% test accuracy. With only 18 test samples the models are very close.



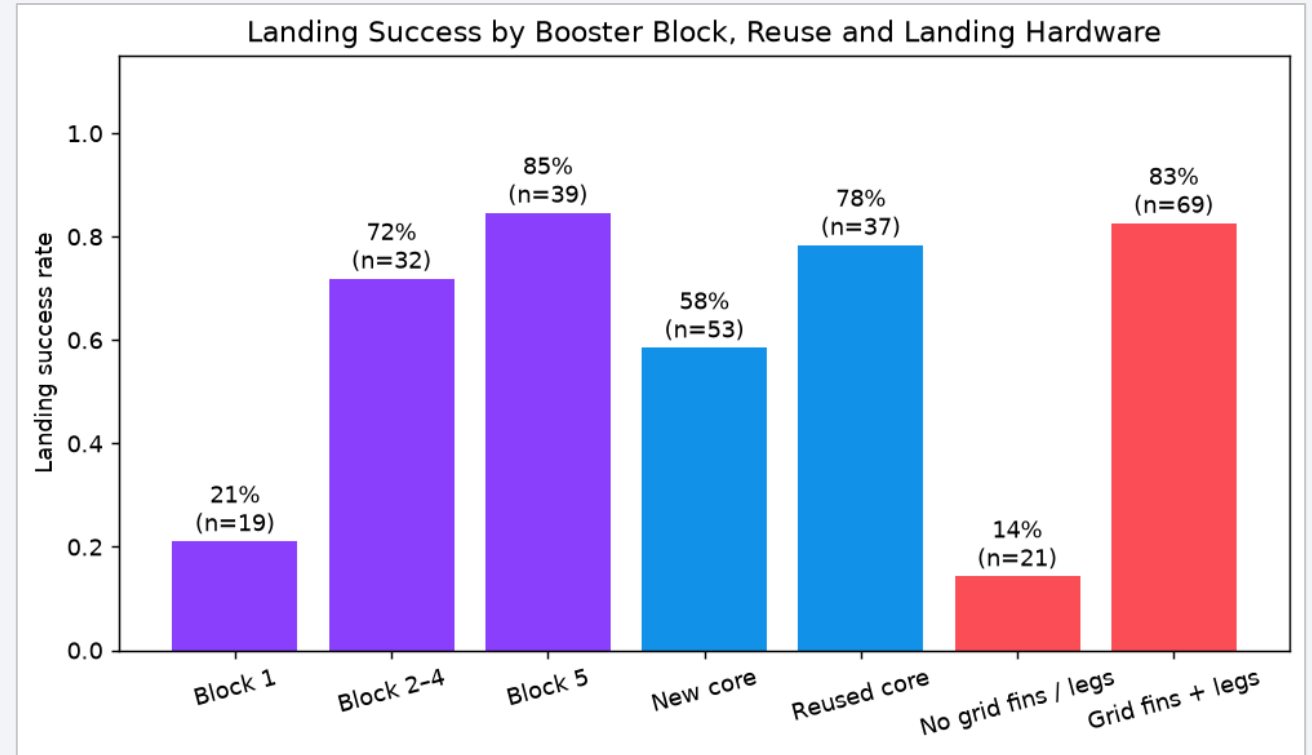
Confusion Matrix

- Confusion matrix of the SVM model on the 18 test samples.
- 12 landings and 3 failures are predicted correctly.
- The model can distinguish between the classes, but its main problem is false positives: 3 launches that did not land are predicted as landed.
- No landed launch is predicted as a failure (no false negatives).



Innovative Insight: Booster Hardware and Reuse

- **Beyond the lab:** we compared the landing success rate by booster Block, core reuse and landing hardware (features already in the data but not analysed in the labs).
- **Landing hardware is decisive:** with grid fins + landing legs 83% landed, without them only 14% (21 launches).
- **Newer boosters land more often:** Block 1 21% → Block 2–4 72% → Block 5 85%.
- **Reuse does not hurt:** reused cores landed 78% of the time vs. 58% for new cores – good news for the cost of reusable launches.
- **For Space Y:** the Block and landing-hardware features should be the first inputs to check when predicting a landing and its cost.



Conclusions

- The landing success rate increases with the number of flights: SpaceX learns from each launch, going from 0% (2010–2013) to over 80% in recent years.
- KSC LC-39A has the most successful launches and the highest success ratio (76.9%).
- Orbits ES-L1, GEO, HEO and SSO have a 100% success rate; GTO is the most difficult frequent orbit (52%).
- Heavy payloads succeed more often in LEO, Polar and ISS orbits; FT and B4 boosters perform best.
- Launch sites are near the coast, highways and railways, and far from cities.
- Landing hardware matters most: boosters with grid fins and landing legs land 83% of the time vs. 14% without them; Block 5 boosters land 85% of the time.
- **Best model:** SVM (sigmoid kernel) with 84.8% cross-validation accuracy and 83.3% test accuracy. It can predict the first-stage landing and help Space Y estimate the cost of a launch.

Appendix

- **GitHub repository:** <https://github.com/luannt0801/pcap-31-03-practice/tree/main/IBM/spacex-capstone>
- **Notebooks and code:**
- **SpaceX API:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/1_data_collection_api.ipynb
- **Web scraping:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/2_web scraping.ipynb
- **Data wrangling:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/3_data_wrangling.ipynb
- **EDA with SQL:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/4_eda_sql.ipynb
- **EDA visualization:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/5_eda_visualization.ipynb
- **Folium map:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/6_folium_launch_site_locations.ipynb
- **Plotly Dash app:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/7_spacex_dash_app.py
- **ML prediction:** https://github.com/luannt0801/pcap-31-03-practice/blob/main/IBM/spacex-capstone/8_machine_learning_prediction.ipynb

```
# Landing label (data wrangling)
bad_outcomes = set(landing_outcomes.keys()[[1, 3, 5, 6, 7]])
df['Class'] = [0 if o in bad_outcomes else 1
               for o in df['Outcome']]

# Model tuning (predictive analysis)
svm_cv = GridSearchCV(SVC(), parameters, cv=10)
svm_cv.fit(X_train, Y_train)
svm_cv.best_params_ # kernel='sigmoid', C=1.0
svm_cv.score(X_test, Y_test) # 0.8333

-- SQL: rank landing outcomes
SELECT Landing_Outcome, COUNT(*) AS Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY Count DESC;
```

Thank you!

